Hyperbolic fun^{n}s.

Circular fun^{n}s.

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$\cos i\underline{x}$.

Hyperbolic fun^{n}s :-

If x is real or complex

$$\cosh x = \frac{e^{x} + e^{-x}}{2}$$

$$\sinh x = \frac{e^{x} - e^{-x}}{2}$$

$\underline{\underline{\sinh ix}}$

$$= \frac{e^{ix} - e^{-ix}}{2}$$

$$= i\left(\frac{e^{ix} - e^{-ix}}{2i}\right)$$

$$= i\sin x$$

Hyperbolic fun^n.

1. Circular fun^n.

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

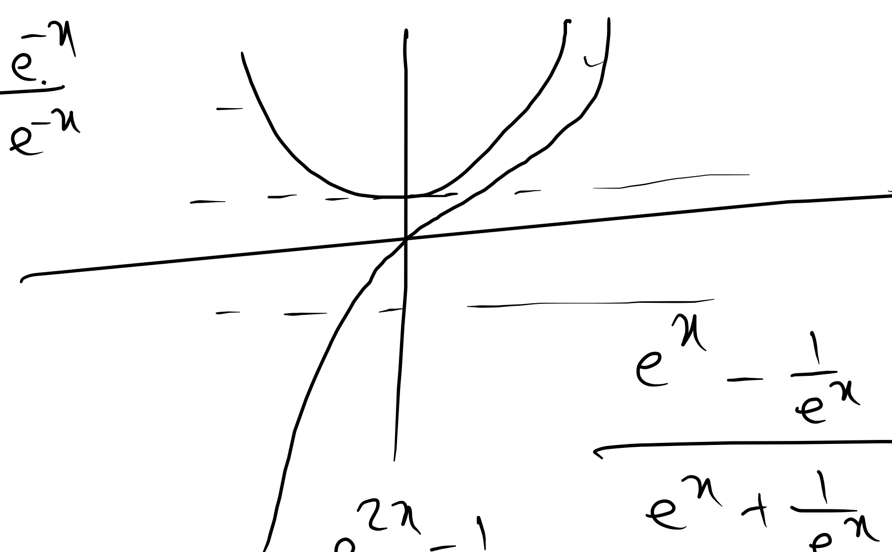
2. Hyperbolic fun^{n}s.

If x is ~~complex~~ real no.
then

$$\cosh x = \frac{e^{x} + e^{-x}}{2}, \quad \sinh x = \frac{e^{x} - e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

x	$-\infty$	0	∞
$\sinh x$	$-\infty$	0	∞
$\cosh x$	∞	1	∞
$\tanh x$	-1	0	1



$$\frac{e^{x} - \frac{1}{e^{x}}}{e^{x} + \frac{1}{e^{x}}}$$

$$\frac{e^{2x} - 1}{e^{2x} + 1}$$

$$= \frac{\cancel{e^{2x}}\left(1 - 1/e^{2x}\right)}{\cancel{e^{2x}}\left(1 + 1/e^{2x}\right)}$$

- If $\tanh x = \frac{1}{2}$, *find* $\sinh 2x$ *and* $\cosh 2x$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{(e^{x} - e^{-x})/\cancel{2}}{(e^{x} + e^{-x})/\cancel{2}} = \frac{1}{2}$$

$$2e^{x} - 2e^{-x} = e^{x} + e^{-x}$$

$$e^{x} - 3e^{-x} = 0$$

$$e^{x} - \frac{3}{e^{x}} = 0 \Rightarrow e^{2x} = 3$$

$$\sinh 2x = \frac{e^{2x} - e^{-2x}}{2}$$

$$= \frac{3 - 1/3}{2} = 4/3$$

$$\cosh 2x = \frac{e^{2x} + e^{-2x}}{2}$$

$$= \frac{3 + 1/3}{2} = 5/3$$

Circular fun^n

$$\cos^2\theta + \sin^2\theta = 1$$
$$\cos^2\theta - \sin^2\theta = \cos 2\theta$$
$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$1 + \cos 2\theta = 2\cos^2\theta$$
$$1 - \cos 2\theta = 2\sin^2\theta$$

$$\sin(A \pm B) = S(A)\,C(B) \pm C(A)\,S(B)$$

$$\cos(A \pm B) = C(A)\,C(B) \mp S(A)\,S(B)$$

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$
$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$
$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$
$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

Hyperbolic fun^n

$$\cosh^2 x - \sinh^2 x = 1$$
$$\cosh^2 x + \sinh^2 x = \cosh 2x$$
$$\sinh 2x = 2\sinh x\cosh x$$

$$1 + \cosh 2x = 2\cosh^2 x$$
$$1 - \cosh 2x = -2\sinh^2 x$$

$$\sinh(A \pm B) = \sinh A\cosh B \pm \cosh A\sinh B$$

$$\cosh(A \pm B) = \cosh A\cosh B \pm \sinh A\sinh B$$

$$2\sinh A\sinh B = \cosh(A+B) - \cosh(A-B)$$

Solve the equation $7\cosh x + 8\sinh x = 1$ for real values of x.

$7\cosh x + 8\sinh x = 1$

$\Rightarrow 7\left(\frac{e^x + e^{-x}}{2}\right) + 8\left(\frac{e^x - e^{-x}}{2}\right) = 1$

$\Rightarrow 7e^x + 7e^{-x} + 8e^x - 8e^{-x} = 2$

$\Rightarrow 15e^x - e^{-x} = 2$

$\Rightarrow 15e^x - \frac{1}{e^x} = 2$

$\Rightarrow 15(e^x)^2 - 1 = 2e^x$

$\Rightarrow 15(e^x)^2 - 2e^x - 1 = 0$

$15y^2 - 2y - 1 = 0$

$e^x = \frac{1}{3}, -\frac{1}{5}$

as x is real $\Rightarrow e^x \neq -1/5$

$\therefore e^x = 1/3 \Rightarrow x = \log(1/3)$

$x = -\log 3$

- Prove that $16\,sinh^5x = \sinh 5x - 5\sinh 3x + 10\sinh x$
- Prove that $16cosh^5x = \cosh 5x + 5\cosh 3x + 10\cosh x$

$$16\sinh^5 x = 16(\sinh x)^5 = 16\left(\frac{e^x - e^{-x}}{2}\right)^5$$

$$= \frac{16}{2^5}\left(e^x - e^{-x}\right)^5$$

$$(a \pm b)^n = a^n \pm {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 \pm \text{———} + b^n$$

$$\begin{array}{ccccccccccccl}
 & & & & & 1 & & & & & & & \\
 & & & & 1 & & 1 & & & & & & \longrightarrow (a+b) \\
 & & & 1 & & 2 & & 1 & & & & & \longrightarrow (a+b)^2 \\
 & & 1 & & 3 & & 3 & & 1 & & & & \longrightarrow (a+b)^3 \\
 & 1 & & 4 & & 6 & & 4 & & 1 & & & \longrightarrow (a+b)^4 \\
1 & & 5 & & 10 & & 10 & & 5 & & 1 & & \longrightarrow (a+b)^5
\end{array}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$16\sinh^5 x = \frac{16}{32}\left(e^x - e^{-x}\right)^5$$

$$= \frac{1}{2}\left[e^{5x} - 5e^{4x}e^{-x} + 10e^{3x}e^{-2x} - 10e^{2x}e^{-3x} + 5e^{x}e^{-4x} - e^{-5x}\right]$$

$$= \frac{1}{2}\left[e^{5x} - e^{-5x} - 5\left(e^{3x} - e^{-3x}\right) + 10\left(e^x - e^{-x}\right)\right]$$

$$= \left(\frac{e^{5x} - e^{-5x}}{2}\right) - 5\left(\frac{e^{3x} - e^{-3x}}{2}\right) + 10\left(\frac{e^x - e^{-x}}{2}\right)$$

$$= \sinh 5x - 5\sinh 3x + 10\sinh x$$

- Prove that
- $\dfrac{1}{1-\dfrac{1}{1-\dfrac{1}{1-cosh^2x}}} = cosh^2x$

$$\cosh^2 x - \sinh^2 x = 1$$
$$\coth^2 x - 1 = \operatorname{cosech}^2 x$$
$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$LHS = \frac{1}{1-\dfrac{1}{1-\dfrac{1}{-\sinh^2 x}}} = \frac{1}{1-\dfrac{1}{1+\operatorname{cosech}^2 x}}$$
$$= \frac{1}{1-\dfrac{1}{\coth^2 x}}$$
$$= \frac{1}{1-\tanh^2 x} = \frac{1}{\operatorname{sech}^2 x}$$
$$= \cosh^2 x$$

- If $u = \log tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)$, Prove that
- **(i)** $\cosh u = \sec\theta$
- **(ii)** $\sinh u = \tan\theta$
- **(iii)** $\tanh u = \sin\theta$
- **(iv)** $\tanh\frac{u}{2} = \tan\frac{\theta}{2}$

$$\Rightarrow e^u = \tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right) = \frac{\tan\pi/4 + \tan\theta/2}{1-\tan\pi/4\,\tan\theta/2}$$

$$e^u = \frac{1+\tan\theta/2}{1-\tan\theta/2}$$

$$\Rightarrow \cosh u = \frac{e^u + e^{-u}}{2} = \frac{1}{2}\left[\frac{1+\tan\theta/2}{1-\tan\theta/2} + \frac{1-\tan\theta/2}{1+\tan\theta/2}\right]$$

$$= \frac{1}{2}\left[\frac{(1+\tan\theta/2)^2 + (1-\tan\theta/2)^2}{1-\tan^2\theta/2}\right]$$

$$= \frac{1}{2}\left[\frac{1+\cancel{2\tan\theta/2}+\tan^2\theta/2 + 1 - \cancel{2\tan\theta/2}+\tan^2\theta/2}{1-\tan^2\theta/2}\right]$$

$$= \frac{1}{\cancel{2}}\left[\frac{\cancel{2}(1+\tan^2\theta/2)}{(1-\tan^2\theta/2)}\right] = \frac{1+\frac{\sin^2\theta/2}{\cos^2\theta/2}}{1-\frac{\sin^2\theta/2}{\cos^2\theta/2}}$$

$$= \frac{(\cos^2\theta/2 + \sin^2\theta/2)\,/\,\cancel{\cos^2\theta/2}}{(\cos^2\theta/2 - \sin^2\theta/2)\,/\,\cancel{\cos^2\theta/2}} = \frac{1}{\cos\theta} = \sec\theta$$

(2)

$$\cosh^2 u - \sinh^2 u = 1$$

$$\Rightarrow \sinh u = \sqrt{\cosh^2 u - 1} = \sqrt{\sec^2\theta - 1} = \tan\theta$$

(3) $$\tanh u = \frac{\sinh u}{\cosh u} = \frac{\tan\theta}{\sec\theta} = \frac{\sin\theta}{\cancel{\cos\theta}}\;\cancel{\cos\theta}$$

(4) $$\tanh u/2 = \frac{\sinh u/2}{\cosh u/2} = \frac{e^{u/2} - e^{-u/2}}{e^{u/2}+e^{-u/2}} = \frac{e^{u/2} - \frac{1}{e^{u/2}}}{e^{u/2} + \frac{1}{e^{u/2}}}$$

$$= \frac{e^u - 1}{e^u + 1}$$

$$= \frac{\frac{1+\tan\theta/2}{1-\tan\theta/2} - 1}{\frac{1+\tan\theta/2}{1-\tan\theta/2} + 1}$$

$$= \frac{\cancel{1} + \tan\theta/2 - (\cancel{1} - \tan\theta/2)}{1 + \cancel{\tan\theta/2} + 1 - \cancel{\tan\theta/2}}$$

$$= \frac{\cancel{2}\tan\theta/2}{\cancel{2}}$$

- If $\cosh x = \sec\theta$, Prove that **(i)** $x = \log(\sec\theta + \tan\theta)$
- **(ii)** $\theta = \frac{\pi}{2} - 2tan^{-1}(e^{-x})$ **(iii)** $\tanh\frac{x}{2} = tan\frac{\theta}{2}$

$\cosh x = \sec\theta \Rightarrow \frac{e^x + e^{-x}}{2} = \sec\theta$

$\Rightarrow e^x + e^{-x} = 2\sec\theta$

$\Rightarrow e^x + \frac{1}{e^x} = 2\sec\theta$

$\Rightarrow (e^x)^2 - 2\sec\theta(e^x) + 1 = 0$

$a = 1, \quad b = -2\sec\theta, \quad c = 1$

$e^x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{+2\sec\theta \pm \sqrt{4\sec^2\theta - 4}}{2}$

$= \frac{2\sec\theta \pm 2\tan\theta}{2}$

$e^x = \sec\theta \pm \tan\theta$

$x = \log(\sec\theta \pm \tan\theta)$

$= \pm\log(\sec\theta + \tan\theta)$ //

$\therefore$ consider $x = \log(\sec\theta + \tan\theta)$

(2) Consider, $\tan^{-1}(e^{-x}) = \alpha$

$\Rightarrow e^{-x} = \tan\alpha$.

$\cosh x = \sec\theta$

$\Rightarrow \frac{e^x + e^{-x}}{2} = \sec\theta \Rightarrow \frac{\cot\alpha + \tan\alpha}{2} = \sec\theta$.

$\Rightarrow \cot\alpha + \tan\alpha = 2\sec\theta$

$\Rightarrow \frac{\cos\alpha}{\sin\alpha} + \frac{\sin\alpha}{\cos\alpha} = 2\sec\theta$

$\Rightarrow \frac{\cos^2\alpha + \sin^2\alpha}{\sin\alpha\cos\alpha} = 2\sec\theta$

$\Rightarrow \frac{1}{2\sin\alpha\cos\alpha} = \frac{1}{\cos\theta}$

$\Rightarrow \frac{1}{\sin 2\alpha} = \frac{1}{\cos\theta}$

$\Rightarrow \sin 2\alpha = \cos\theta$

$\Rightarrow \cos\left(\frac{\pi}{2} - 2\alpha\right) = \cos\theta$

$\frac{\pi}{2} - 2\alpha = \theta$

$\Rightarrow \theta = \pi/2 - 2\tan^{-1}(e^{-x})$

(3) $\tanh\frac{x}{2}$

Relation betn circular & Hyperbolic funcns

$\sin(ix) = i\sinh x$

$\cos ix = \cosh x$

$\sinh ix = i\sin x$

$\cosh ix = \cos x$

$\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$

$\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$

$\dfrac{e^{-x} - e^{x}}{2i}$

$-i\left(\dfrac{e^{-x} - e^{x}}{2}\right)$

$\cosh\theta = \dfrac{e^{\theta} + e^{-\theta}}{2}$

$\sinh\theta = \dfrac{e^{\theta} - e^{-\theta}}{2}$

$\cos ix = \dfrac{e^{-x} + e^{x}}{2}$

$i\left(\dfrac{e^{x} - e^{-x}}{2}\right)$

Separation into real & imag. parts

$\sin ix = i \sinh x \qquad \cos ix = \cosh x$

$\sinh ix = i \sin x \qquad \cosh ix = \cos x$

$$\sin(x \pm iy) = \sin x \cos iy \pm \cos x \sin iy$$
$$= \sin x \cosh y \pm i \cos x \sinh y$$

$$\cos(x \pm iy) = \cos x \cos iy \mp \sin x \sin iy \quad \text{HW}$$
$$=$$

$$\cosh(x \pm iy) = \cosh x \cosh iy \pm \sinh x \sinh iy$$
$$= \cosh x \cos y \pm i \sinh x \sin y$$

$\sinh(x \pm iy) =$ HW

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given $\tan(x+iy) = \alpha + i\beta$ To find x & y

$\rightarrow \tan(x-iy) = \alpha - i\beta$

$$\tan 2x = \tan\Big(\underbrace{x+iy}_{A} + \underbrace{x-iy}_{B}\Big) = \frac{\tan(x+iy) + \tan(x-iy)}{1-\tan(x+iy)\tan(x-iy)}$$

$$\tan 2x = \frac{\alpha + \cancel{i\beta} + \alpha - \cancel{i\beta}}{1-(\alpha+i\beta)(\alpha-i\beta)} = \frac{2\alpha}{1-(\alpha^2+\beta^2)}$$

$$\tan(i2y) = \tan\big(x+iy - (x-iy)\big) = \frac{\tan(x+iy) - \tan(x-iy)}{1+\tan(x+iy)\tan(x-iy)}$$

$$\cancel{i}\tanh 2y = \frac{\alpha+i\beta-(\alpha-i\beta)}{1+(\alpha+i\beta)(\alpha-i\beta)} = \frac{\cancel{i}2\beta}{1+\alpha^2+\beta^2}$$

- Separate into real and imaginary parts $tan^{-1}(e^{i\theta})$

$$\tan^{-1}(e^{i\theta}) = x + iy$$
$$e^{i\theta} = \tan(x+iy)$$

$$\tan(x+iy) = \cos\theta + i\sin\theta$$

To find x & y

$$\tan(x-iy) = \cos\theta - i\sin\theta$$

$$2x = x+iy + x-iy$$
$$\tan(2x) = \tan[(x+iy) + (x-iy)]$$
$$= \frac{\tan(x+iy) + \tan(x-iy)}{1 - \tan(x+iy)\tan(x-iy)}$$
$$= \frac{\cos\theta + \cancel{i\sin\theta} + \cos\theta - \cancel{i\sin\theta}}{1 - (\cos^2\theta + \sin^2\theta)}$$
$$= \frac{2\cos\theta}{1-1} = \infty$$
$$\Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \pi/4$$

$$2iy = (x+iy) - (x-iy)$$
$$\tan(2iy) = \tan[(x+iy) - (x-iy)]$$
$$= \frac{\tan(x+iy) - \tan(x-iy)}{1 + \tan(x+iy)\cdot\tan(x-iy)}$$
$$= \frac{\cancel{\cos\theta} + i\sin\theta - (\cancel{\cos\theta} - i\sin\theta)}{1+1}$$

$$\cancel{i}\tanh 2y = \frac{\cancel{2}i\sin\theta}{\cancel{2}} \Rightarrow 2y = \tanh^{-1}(\sin\theta)$$
$$\Rightarrow y = \frac{1}{2}\tanh^{-1}(\sin\theta)$$

- If $\sin(\alpha - i\,\beta) = x + i\,y$ then prove that $\frac{x^2}{cosh^2\beta} + \frac{y^2}{sinh^2\beta} = 1$ and $\frac{x^2}{sin^2\alpha} - \frac{y^2}{cos^2\alpha} = 1$

$$\sin(\alpha - i\beta) = x + iy$$

$$\sin\alpha \cos i\beta - \cos\alpha \sin i\beta = x + iy$$

$$\sin\alpha \cosh\beta - i\cos\alpha \sinh\beta = x + iy$$

$$x = \sin\alpha \cosh\beta \qquad y = -\cos\alpha \sinh\beta$$

HW

- If $cos(x + i\,y) = \cos\alpha + i\sin\alpha$ *, prove that*
- **(i)** $\sin\alpha = \pm sin^2 x = \pm \sin h^2 y$
- **(ii)** $\cos 2x + \cosh 2y = 2$

$\cos(x+iy) = \cos\alpha + i\sin\alpha$

$\cos x \cos iy - \sin x \sin iy = \cos\alpha + i\sin\alpha$

$\cos x \cosh y - i\sin x \sinh y = \cos\alpha + i\sin\alpha$

$\cos x \cosh y = \cos\alpha, \quad -\sin x \sinh y = \sin\alpha$ — (1)

$\cosh y = \frac{\cos\alpha}{\cos x} \qquad \sinh y = -\frac{\sin\alpha}{\sin x}$

$\cosh^2 y - \sinh^2 y = 1$

$\frac{\cos^2\alpha}{\cos^2 x} - \frac{\sin^2\alpha}{\sin^2 x} = 1$

$\Rightarrow \cos^2\alpha \sin^2 x - \sin^2\alpha\cos^2 x = \cos^2 x \sin^2 x$

$\Rightarrow (1-\sin^2\alpha)\sin^2 x - \sin^2\alpha(1-\sin^2 x) = (1-\sin^2 x)\sin^2 x$

$\Rightarrow \cancel{\sin^2 x} - \cancel{\sin^2\alpha\sin^2 x} - \sin^2\alpha + \cancel{\sin^2\alpha\sin^2 x} = \cancel{\sin^2 x} - \sin^4 x$

$\Rightarrow \sin^2\alpha = \sin^4 x \Rightarrow \sin\alpha = \pm\sin^2 x$

T.P. $\sin\alpha = \pm\sinh^2 y$

from (1) $\cos x = \frac{\cos\alpha}{\cosh y} \qquad \sin x = -\frac{\sin\alpha}{\sinh y}$

$\cos^2 x + \sin^2 x = 1$

$\frac{\cos^2\alpha}{\cosh^2 y} - \frac{\sin^2\alpha}{\sinh^2 y} = 1$

$\cos^2\alpha = (1-\sin^2\alpha$

$\cosh^2 y = 1+\sinh^2 y$

$\sin\alpha = \pm\sinh^2 y$

(2) T.P.T.

$\cos 2x + \cosh 2y = 2$

$1+\cos 2x = 2\cos^2 x$

$1-\cos 2x = 2\sin^2 x$

$1+\cosh 2x = 2\cosh^2 x$

$1-\cosh 2x = -2\sinh^2 x$

$\cos 2x + \cosh 2y$

$= 1 - 2\sin^2 x + 1 + 2\sinh^2 y$

$= 1 - 2\cancel{\sin\alpha} + 1 + 2\cancel{\sin\alpha}$

$= 2$

$\sin\alpha = \pm\sin^2 x = \pm\sinh^2 y$

- If $x + i\,y = \tan(\pi/6 + i\,\alpha)$, prove that $x^2 + y^2 + 2x/\sqrt{3} = 1$
- **Solution:**

$$\tan(\pi/6 + i\alpha) = x + iy$$

$$\tan(\pi/6 - i\alpha) = x - iy$$

$$\frac{2\pi}{6} = \pi/6 + i\alpha + \pi/6 - i\alpha$$

$$\tan(\pi/3) = \tan\left((\pi/6 + i\alpha) + (\pi/6 - i\alpha)\right)$$

$$= \frac{\tan(\pi/6 + i\alpha) + \tan(\pi/6 - i\alpha)}{1 - \tan(\pi/6 + i\alpha)\tan(\pi/6 - i\alpha)}$$

$$\sqrt{3} = \frac{(x + \cancel{iy} + x - \cancel{iy})}{1 - (x+iy)(x-iy)} \Rightarrow \sqrt{3} = \frac{2x}{1 - (x^2 + y^2)}$$

$$1 - (x^2 + y^2) = \frac{2x}{\sqrt{3}} \Rightarrow x^2 + y^2 + \frac{2x}{\sqrt{3}} = 1$$

If $u + i\,v = cosec\left(\frac{\pi}{4} + i\,x\right)$, *prove that* $(u^2 + v^2)^2 = 2(u^2 - v^2)$

$$\frac{1}{\sin(\pi/4 + ix)} = u + iv$$

$$\sin(\pi/4 + ix) = \frac{1}{u+iv} \times \frac{u-iv}{u-iv}$$

$$\sin\pi/4 \cos ix + \cos\pi/4 \sin ix = \frac{u-iv}{u^2+v^2}$$

$$\frac{1}{\sqrt{2}} \cosh x + i\frac{1}{\sqrt{2}} \sinh x = \frac{u}{u^2+v^2} - i\frac{v}{u^2+v^2}$$

$$\frac{\cosh x}{\sqrt{2}} = \frac{u}{u^2+v^2} \qquad \frac{\sinh x}{\sqrt{2}} = \frac{-v}{u^2+v^2}$$

$$\cosh x = \frac{\sqrt{2}\,u}{u^2+v^2} \qquad \sinh x = \frac{-\sqrt{2}\,v}{u^2+v^2}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\frac{2u^2}{(u^2+v^2)^2} - \frac{2v^2}{(u^2+v^2)^2} = 1 \Rightarrow 2(u^2-v^2) = (u^2+v^2)^2$$